ENGINEER 1P13:
PROJECT THREE WORKSHEETS (INDIVIDUAL)

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PROJECT THREE MILESTONE ONE: PROBLEM FRAMING AND TEST PLAN

MILESTONE 1.1 – CLIENT NOTES

Team ID: Mon-23

Complete this worksheet individually before coming to Design Studio for Week 6.

1. Include your client notes from the introductory client visit

Name: Prabhjot Bhandal	MacID: bhandp9

Client Profile:

Name: Kimberly

Age: About 60, Right Handed

Interests: Food, Hiking, Playing Music and Dancing (**Self-Actualization, Safety Needs**)

- Likes playing guitar

Disability: Usher's Syndrome, it slowly deteriorates your hearing, vision and balance as you get older.

- Uses a walking cane (like the one blind people use), **affects physical activity**
- Uses a hearing aid, **affects listening and comprehension**
- No allergies
- Likes bright colours, needs contrast
- CAN'T SEE COLOUR, COLOURBLIND
- Brings her charger with her, especially with Bluetooth
- iPhone 12 Pro Max user (eww….)

Notes:

Tell us about yourself

- Usher's syndrome,
- Born in the 60's, lots of viruses around then so she found out late
- Got cataracts which made it worse (clouding of the lens)
- Audio helps her the most, has the most trouble with vision but has easier time with audio
- All accessibility is just beeping and beeping often doesn't work for her, it hurts her ears because of her hearing aids
- Counts her steps to see how far things are
- She outlines objects with her cane to see what it is, but only does that with her phone
- Needs to magnify objects and words to see what she is looking at/dealing with
- Lives with her older son but he works a lot

    - She meets with her intervener once a week, someone who teaches her how to get around
- Doesn't have an interest to read braille, she can't really feel the tactile dots
- Fast Learner

Guitar/Music

- Knows the basics
- Comfortable playing guitar itself and easy with chords, HARD TO READ THE NOTES
    - Never can read the notes, needs to "feel" the notes
- Likes anything but electronic music because of the hearing aids, very susceptible to high frequencies
- Likes karaoke
- Large group of instruments is hard
- Does like listening to instrumentals and songs with lyrics
    - Would like to learn instrumental first than couple months down would learn the lyrics and sing

Dancing

- Usually pre-learns the dance so that she can keep up with the other dancers
- Would like to get into different types of dancing

Wishes

- Work together, collaborate
- For her, she needs to be hands-free so that she can use her hands for other things
- Like to have verbal audio descriptions, likes to have hands free
- Likes analog/manual control, kinda like a knob or buttons

Apps on Phone

- High contrast, but then updates can mess up the formatting and settings, then has to relearn where everything is

Transportation

- Can lose a sense of direction and location because she is never told where she is by her accessibility apps
- She responds best to verbal/audio
- Someone often helps her go to new places
- Phone doesn't tell her what she is stepping on/surroundings, bad for snow and ice
- She feels like an easy target outside and gets disoriented relatively often, doesn't wanna be taken advantage of
- Often tags along with people

Cooking

- Memorizes the location of her pots and pans when cooking to prevent hurting herself
- Can't cut food by hand and is scared to do so, she wants to cut it herself but gets it from the store instead
- Wants something to tell her what she is holding, object recognition

- **Favourite Food:** Everything
- Can't see oil in the pan, can't panfry
- Doesn't care about the look of food

Hiking

- Really wants to go to the waterfalls, most hiking areas have lots of obstacles

Questions For Interests:

1. What's the one thing you miss most as you've gotten older and why?

2. I've heard that you like to play guitar. How has your ability to play guitar been affected by your disability?

3. What do you like the most about hiking?

4. What's your favorite app on your phone and why do you like that app so much?
   Really likes using Bluetooth for audio

Questions for Needs:

1. What's something that you find really tedious to do in your everyday life?

Tools:

- High contrast, bright-colours to help her see shapes and boundaries
  - Often uses her phone to help with contrast
- Verbal audio that is easy to understand
- Camera:
  - Uses AI first (Seen AI is the app, describes in lots of details)
  - Zoom In/Magnifier
  - Takes a picture
- Really likes using Bluetooth
- Wants descriptions
- Siri is alright, no challenges with Siri as of now
- Can wear something along the neck, but it needs to be light
  - Prefer heavier wearables on the front
- Doesn't care about the look of the solution

PROJECT THREE MILESTONE TWO: DESIGN EXPLORATION AND DESIGN REVIEW #1

MILESTONE 2.1 – RESEARCH ASSIGNMENT

Team ID: Mon-23

Complete this worksheet before Design Studio for Week 7.

- State the question you plan to answer through your research
- Summarize your research findings (answer). Your answer should be a coherent, well-written summary of your research, not a "brain dump".
- You may include images, but don't forget to cite them properly.
- Aim for a length of about 500 words
- Properly cite your sources using IEEE formatted references and in-text citations. For information on referencing formats and choosing sources, see Design and Communication Workshop 1.

Name: Prabhjot Bhandal	MacID: bhandp9

What is your question?

Do previously able-bodied individuals feel less connected to the world after acquiring a disability? What strategies help a previously able-bodied individual adapt to reduced spatial awareness caused by a disability?

What is your answer?

When someone experiences reduced spatial awareness due to a recently acquired disability, it can lead to a feeling of disconnection from the world. This is very true for individuals with disabilities as they are less likely to have close friends, relatives and local contacts [1,2]. Moreover, the loss of spatial awareness makes it more difficult for them to experience positive social interactions, which worsens their mental health [1,2]. To help individuals adapt and navigate their reduced spatial awareness, several strategies are used, including environment modification, sensory integration and mental representation exercises [3,4].

One strategy that is used is to modify the individual's everyday environment. By rearranging furniture or creating wider pathways, individuals can make their homes easier and safer to navigate [3]. Visual cues, like bright colours and distinct lighting, can be used to identify separate locations within a space [3]. Bright lighting can also aid in this by increasing contrast but can also reduce potential hazards and make different objects easier to see [3].

Another strategy that is used involves utilizing all sensory information to compensate for spatial limitations. People with limited spatial awareness can develop techniques to use all their available

senses to understand their surroundings [4]. For example, mobility aids like white canes allow for visually impaired individuals to interpret their environment through tactile feedback and sound, which helps them regain some of the spatial awareness they have lost [5]. This recontextualization of sensory input enables them to navigate effectively without having to only rely on sight [5].

Alternatively, individuals can engage in mental representation exercises, which help improve spatial awareness and fine motor skills [4,6]. Through motor imagery training, individuals mentally simulate physical movements, improving motor planning, coordination, and precision, which are all crucial for those with limited spatial awareness [7]. Basic Action Concepts (BACs) are an extension of motor imagery as they help improve cognitive abilities related to performing complex actions with precision [6,7]. These exercises enable individuals to enhance their spatial awareness and fine-tune their movements without relying on external aids [4,6].

In conclusion, there are various ways to improve spatial awareness and reconnect individuals with their environment. Environmental modifications help reduce hazards and make navigation easier. By utilizing other senses in intuitive ways, with or without aids, individuals can better understand their surroundings. Alternatively, mental exercises can actively improve spatial awareness, enhancing both motor skills and precision. Together, these strategies can help individuals regain a strong sense of connection to their environment.

List of sources:

[1] Y. Strautins and C. McDiarmid, "Social connections among persons with and without mental health-related disabilities, 2020," *www150.statcan.gc.ca*, May 31, 2023. https://www150.statcan.gc.ca/n1/pub/89-654-x/89-654-x2023001-eng.htm

[2] Statistics Canada, "The Daily — Exploring the social interactions of those with and without mental health-related disabilities," *Statcan.gc.ca*, 2023. https://www150.statcan.gc.ca/n1/daily-quotidien/230531/dq230531f-eng.htm

[3] The OT Centre UK, "Private Occupational Therapy and Sensory Integration | The OT Centre - Spatial Awareness Deficits," *Theotcentre.co.uk*, 2024. https://www.theotcentre.co.uk/difficulties-treated-with-occupational-therapy-and-si-therapy/spatial-awareness-deficits

[4] J. Duquette and F. Baril, "Spatial orientation in adolescents with visual impairment: related factors and avenues for assessment," Mar. 2012. Available: https://extranet.inlb.qc.ca/wp-content/uploads/2015/01/Spatial-orientation-in-adolescents-with-VI_factors_assessment.pdf

[5] CNIB Foundation, "About the White Cane," *CNIB*. https://www.cnib.ca/en/about-white-cane?region=on

[6] T. Schack, K. Essig, C. Frank, and D. Koester, "Mental representation and motor imagery training," *Frontiers in Human Neuroscience*, vol. 8, May 2014, doi: https://doi.org/10.3389/fnhum.2014.00328.

[7] M. F. Almufareh, S. Kausar, M. Humayun, and S. Tehsin, "Leveraging Motor Imagery Rehabilitation for Individuals with Disabilities: A Review," *Healthcare*, vol. 11, no. 19, p. 2653, Jan. 2023, doi: https://doi.org/10.3390/healthcare11192653.

MILESTONE 2.1 – INITIAL CONCEPT EXPLORATION

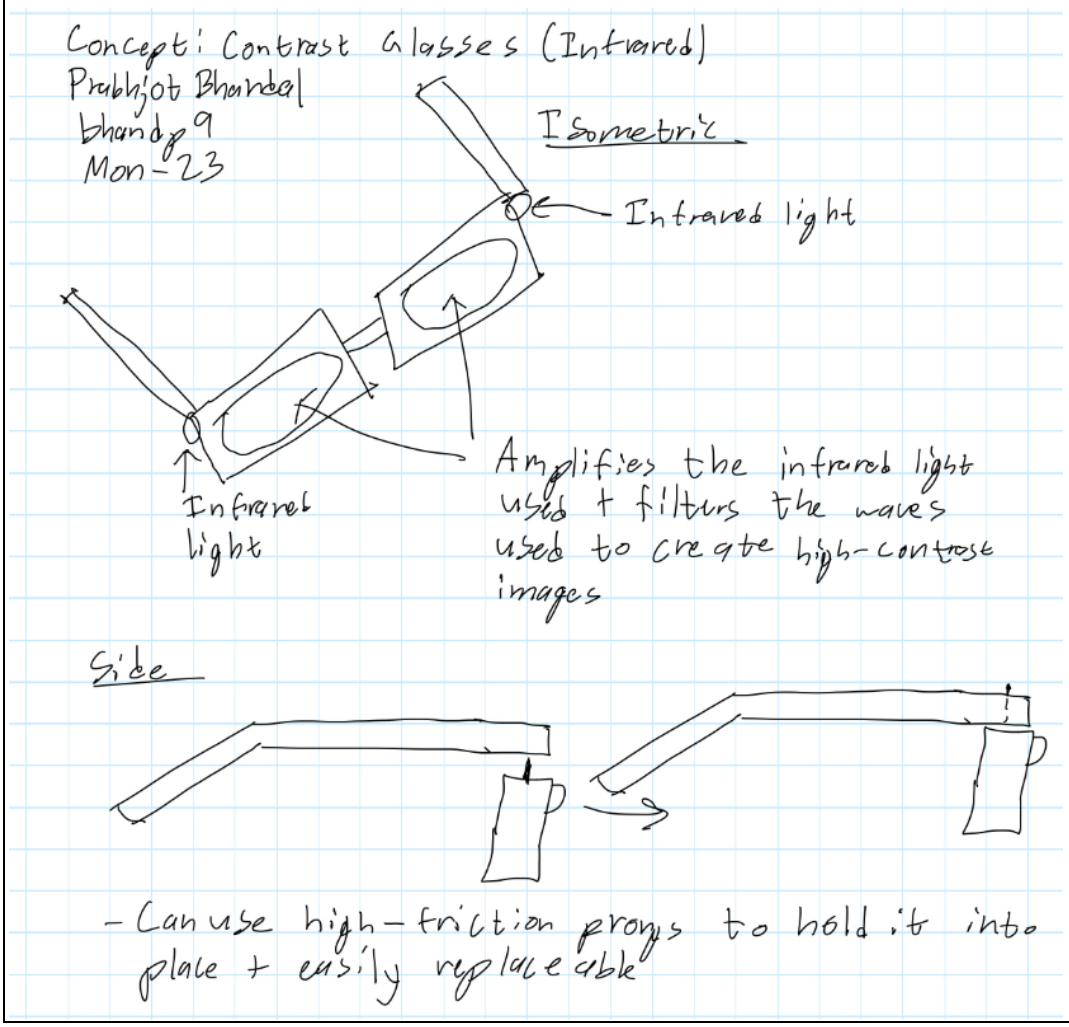
Team ID: Mon-23

Complete this worksheet before Design Studio for Week 7.

1. Include multiple images of your **initial** concept exploration, if needed
   → Include necessary annotations to help in the communication of your ideas
   → These can be photos of hand sketches, photos of initial prototypes, screen grabs of basic CAD models
   → Include your Team Number, Name and MacID on *each* concept image
2. Insert your photo(s) as a Picture (Insert > Picture > This Device)
3. **Do not include more than *two* concept images per page**

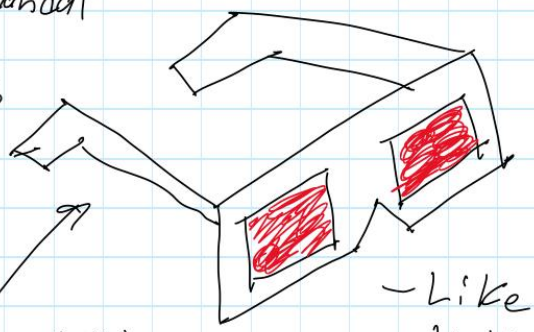
Name: Prabhjot Bhandal	MacID: bhandp9

Insert screenshot(s) of your concept below.



Alternative: Contrast Glasses that work with opposite colours.

Prabhjot Bhandal
bhandp9
Mon-23



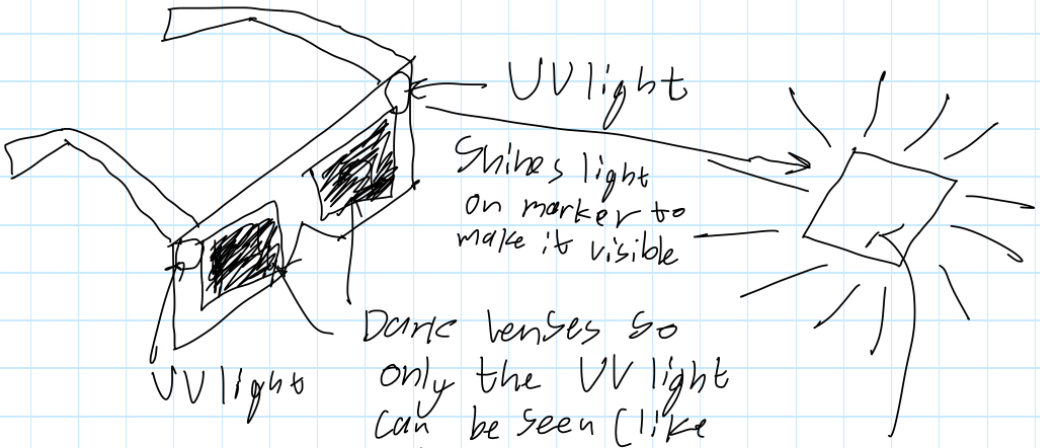


would be cost-effective ; 3D glasses at movie theaters are cheap & easy to make

- Like 3D glasses, can distinctly see objs of opposite colour
  ↳ issue: would need to get lots of Live objects

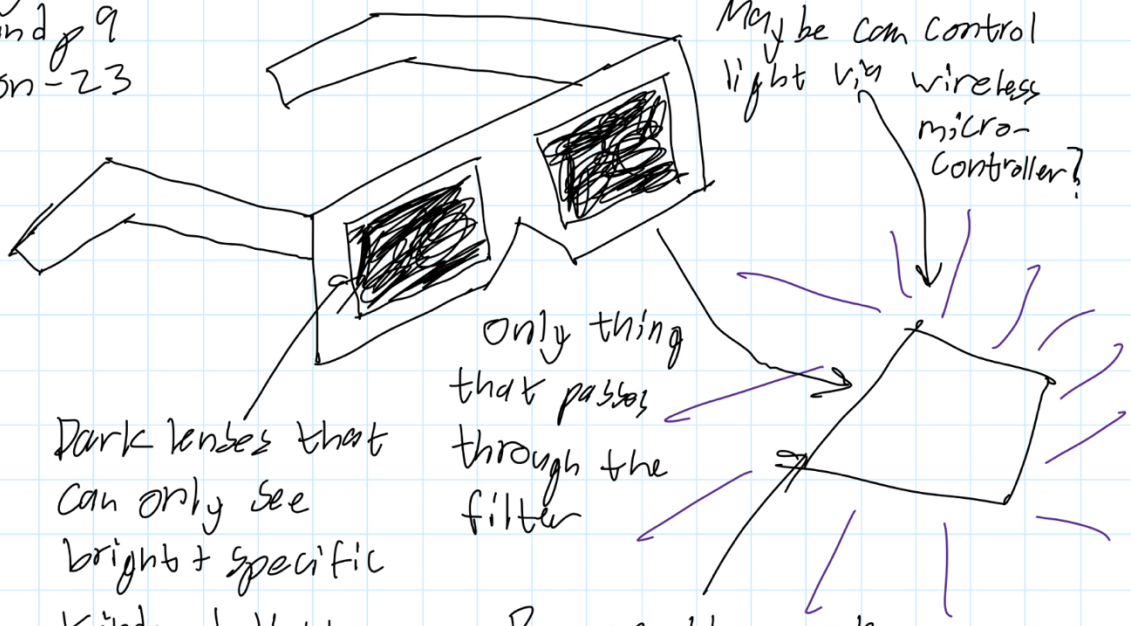
Alternative: Contrast glasses w/ UV light and reuseable markers.

Prabhjot Bhandal
bhandp9
Mon-23

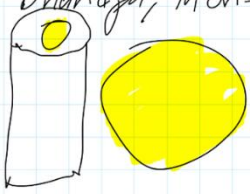


Alternative! Contrast Glasses w/ lenses that point out ~~glowing~~ markers

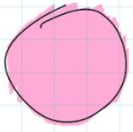
Prabhjot Bhandal
bhandp9
Mon-23



Concept: Audio Sticky Notes
Prabhjot Bhandal,
bhandp9, Mon-23



- Sticks onto anything
- Has a programmable NFC tag
  - ↳ Can be programmed w/ a phone
  - ↳ Text-to-speech audio
- Heat resistant
- High-contrast, neon colour
- Reusable



← can come in diff. colours

Issues:
- Finite
- would need to know what something is before labelling
- Isn't obvious how to use it immediately

Fixes?!
- Make a QR code generator, infinite QR codes
- Give her prompts of how to use it
  - ↳ would be in audio

MILESTONE 2.3 – REFINED CONCEPT EXPLORATION

Team ID: Mon-23

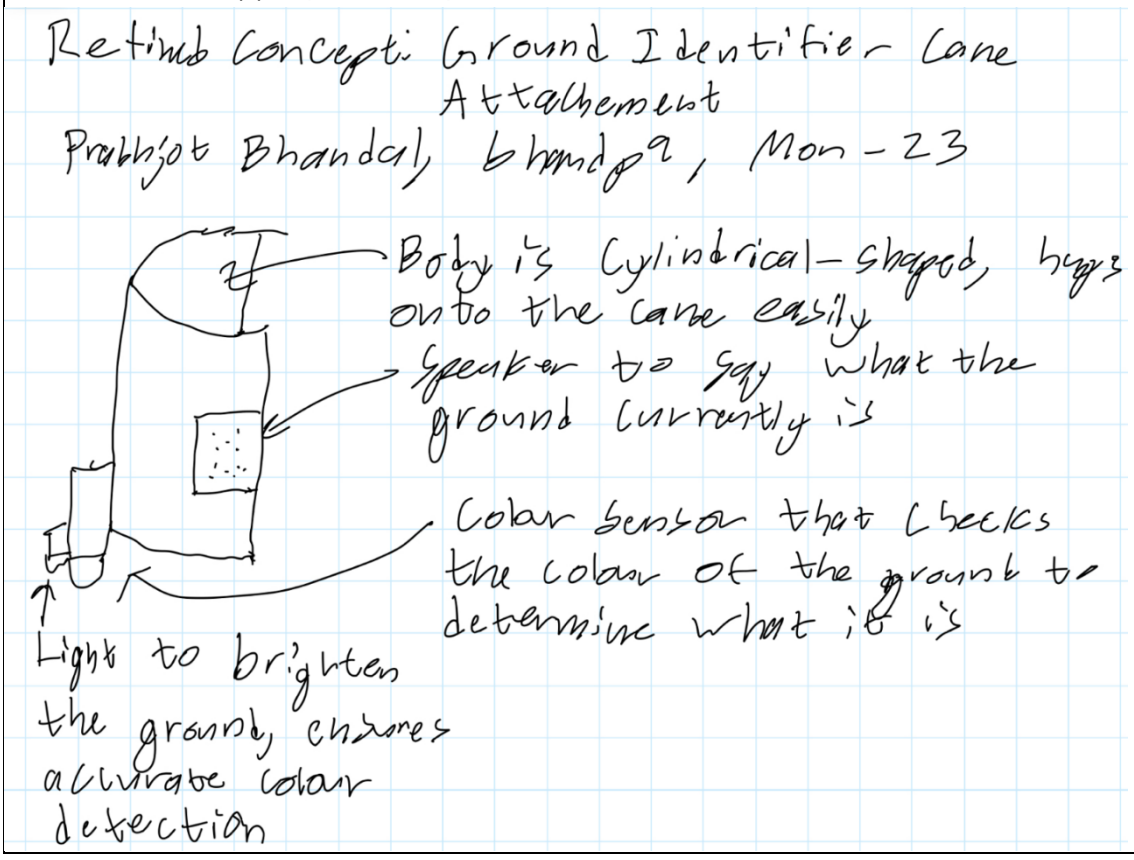
Complete this worksheet <u>during</u> Design Studio for Week 7.

4. Include multiple images of your **refined** concept exploration, if needed
   - → Include 2 distinct concepts based on the functional analysis
   - → Include necessary annotations to help in the communication of your ideas
   - → These can be photos of hand sketches, photos of initial prototypes, screen grabs of basic CAD models
   - → Include your Team Number, Name and MacID on <u>*each*</u> concept image
5. Insert your photo(s) as a Picture (Insert > Picture > This Device)
6. **<u>Do not include more than *two* concept images per page</u>**

Concept 1: Ground Identifier Cane Attachment

Name: Prabhjot Bhandal	MacID: bhandp9

Insert screenshot(s) of your concept below.

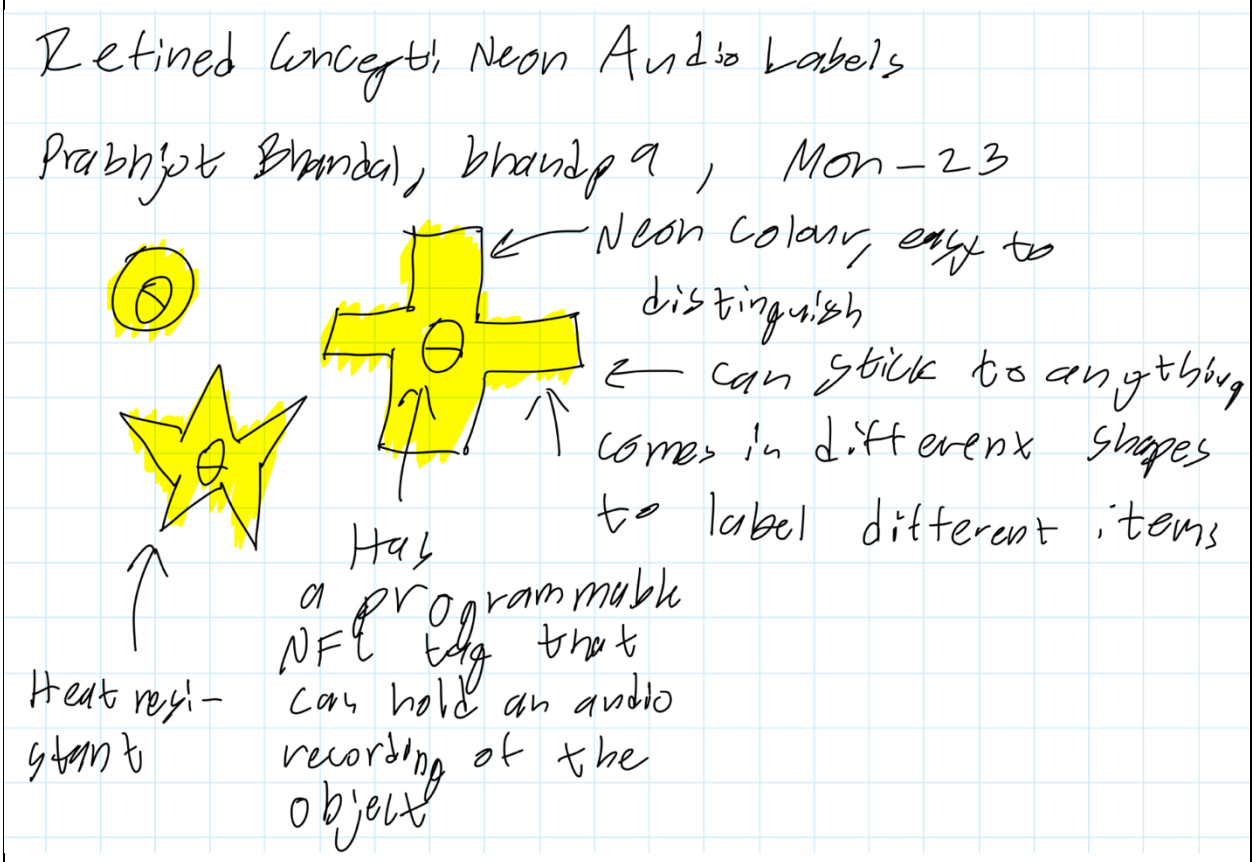


Team ID: Mon-23

Concept 2: Neon Audio Labels in Fun Shapes

Name: Prabhjot Bhandal	MacID: bhandp9

Insert screenshot(s) of your concept below.



PROJECT THREE MILESTONE THREE: PROTOTYPING, DECISION MAKING AND DESIGN REVIEW #2

MILESTONE 3.1 – REFINED CONCEPT: INITIAL PROTOTYPE

Team ID: Mon-23

Complete this worksheet individually before coming to Design Studio/Lab A for Week 8.

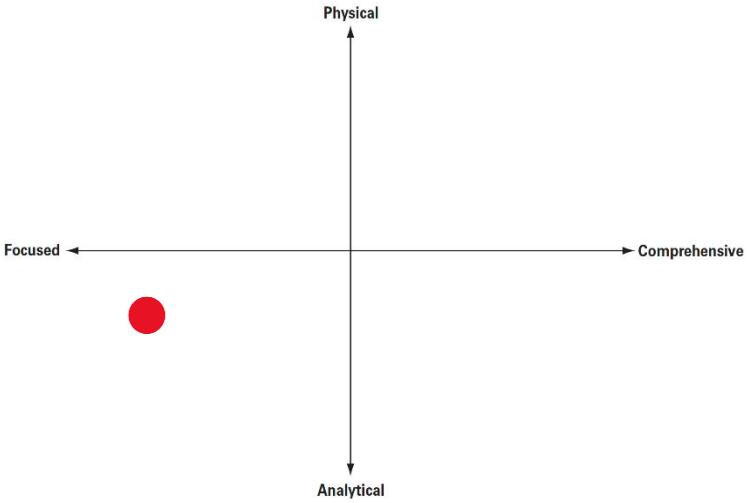
1. Write a small description of your initial prototype. Be sure to include what problem it aims to solve, how your initial prototype will be fabricated, and what functionality will be included and omitted in this initial prototype.
2. Classify whether your prototype is Physical or Analytical, and Focused or Comprehensive. Include the purpose of this prototype in the context of project 4 and the level of fidelity (low, medium, or high fidelity)
   - → **Physical vs. Analytical:** Physical prototypes are tangible artifacts that are created to approximate the final product. Analytical prototypes are non-tangible and represent the product using usually visual or mathematical models.
   - → **Focused vs. Comprehensive:** Focused prototypes implement only one or a few of the attributes of the final product. Comprehensive prototypes aim to implement most, if not all of the attributes of the final product.
3. Create a list of objectives and metrics for your initial prototype. There is no required amount of objectives or metrics, so long as the list is comprehensive.
4. Create a rough experimental plan for how you might test your prototype. Consider the methods you might use to test various objectives, how you will measure how effective each test proves to be and how realistic it would be to implement. This does not need to be detailed plan but should consider several of your objectives for the prototype.
5. Take picture(s) of your refined concept (initial prototype)
   - → Insert your photo(s) as a Picture (Insert > Picture > This Device)
   - → **Do not include more than *two* refined concept pictures per page**
   - → Include details on how concept was refined (what feedback was incorporated, what features are different than previous concept exploration, etc.)
   - → You can continue this process within the allocated time of the LabA/DS and seek feedback and discussions from your team members and/or the instructional team (IAIs, TAs, etc.).

Name: Prabhjot Bhandal	MacID: bhandp9

Write a short description of your initial prototype below.

The sidewalk indent detector utilizes the detection of the joints in sidewalks to determine if the user is currently walking on a sidewalk. This prototype is built using an Arduino microcontroller, a breadboard, 3 LEDs and a photoresistor, which acts as a stand-in for an IR/colour sensor. A white LED will be used to generate some light for the IR/colour sensor to recognize areas with high or low contrast, with low contrast areas representing the indents in the sidewalk. When an indent is detected, an orange LED blinks to show that the indent was detected. If an indent has not been found for quite some time (4 seconds), a blue LED turns on, signaling that the user is no longer moving on a sidewalk and may be in danger.

Indicate where your prototype falls on the scale below.



Kind of Prototype:

☐ Physical or ☐ Analytical

☐ Focused or ☐ Comprehensive

Purpose of Prototype: To determine whether the user is walking in a safe or dangerous area.

Level of Fidelity: Low to Medium

Include a list of objectives and metrics for your prototype below.

Objectives

- Provides effective feedback to the user when they are not walking on a sidewalk
- Accurately detects and identifies indents, even in low-light conditions
- Determines if the user is walking on a sidewalk or not

Metrics

- Time it takes the user to notice the alarm (seconds)
- Number of Missed Indents
- Number of False Alarms

Include a rough experimental plan on how you might test your prototype below.

Objective: Provides effective feedback to the user when they are not walking on a sidewalk

Metric: Time it takes the user to notice the alarm (seconds)

Test: Conduct 3 sets of 5-7 trials during the day, night with streetlights and night without streetlights. Each set would vary the LED's light intensity and the reaction time of the user to notice

the alarm will be recorded. The light intensity with the lowest average reaction time will be the best one.

Objective: Accurately detects and identifies indents, even in low-light conditions

Metric: Number of Missed Indents

Test:

Conduct 3-4 sets of 4 trials with each set testing a different sight LED (the white LED that is used to create high-contrast conditions for the IR/colour sensor) angle. Each trial would walk over a predetermined path with known indent locations. The missed indents per trial would be recorded and each angle's missed indent average will be determined. The most optimal sight LED angle will be based on the lowest average of missed indents.

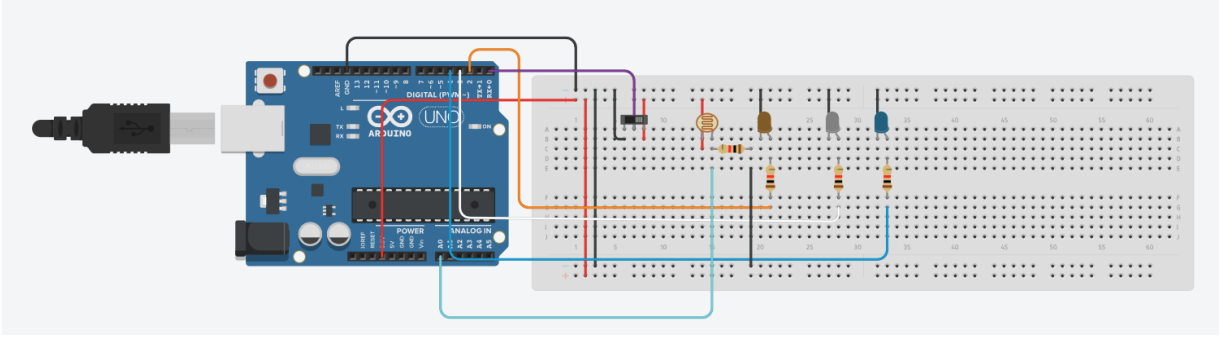
Objective: Determines if the user is walking on a sidewalk or not

Metric: Number of False Alarms

Test: Conduct 3 sets of 5 trials, each with a different time threshold. Each trial will be conducted on and off the sidewalk at a consistent speed. The number of false alarms would be measured per trial and an average would be calculated for each different time threshold. The time threshold with the lowest false alarm rate would be chosen.

Insert picture(s) of your refined concept (initial prototype) below.

Interfacing:



Code:

//NAME: Prabhjot Bhandal

//DATE: March 6th, 2025

//TITLE: 1P13 P3 MS3: Indent Investigator

```
//VARIABLES
//BUTTONS + LEDS
int START_SWITCH = 0; //PURPLE
int INDENT_DETECTED_LED = 2; //ORANGE
int CONTRAST_LED = 3; //WHITE
int NOT_ON_SIDEWALK_LED = 4; //BLUE

//SENSORS
//PHOTORESISTOR, stand in for a colour sensor (contrast),
//assigned to pin A0, analog!!!
// <450 = Dark, Indent Detected
// >450 = Light, Sidewalk Detected

//LOGIC VARIABLES
//unsigned long is a data type that keeps its entire value
//and can only be positive.
unsigned long lastIndentTime = 0;



void setup()
{
  //START BUTTON
  pinMode(START_SWITCH, INPUT);
  //CONTRAST
  //NOTE: For interfacing Photoresistor, must connect another
  //resistor to it to ground it.
  pinMode(A0, INPUT);

  //LEDS
```

```
  pinMode(INDENT_DETECTED_LED, OUTPUT);
  pinMode(CONTRAST_LED, OUTPUT);
  pinMode(NOT_ON_SIDEWALK_LED, OUTPUT);

  //Starts up serial monitor for comments
  Serial.begin(9600);
}

void loop()
{
  //Starts internal timer, how long it has been running
  unsigned long currentTime = millis();
  Serial.println(analogRead(A0));

  //Once switch activated, circuit turns on
  if (digitalRead(START_SWITCH) == HIGH) {
    //Turn on sight LED
    digitalWrite(CONTRAST_LED, HIGH);

    //When indent detected
    if (analogRead(A0) < 325) {
      //Indent detected
      Serial.println("Indent Detected; Still on Sidewalk");

      //Reset Timer
      lastIndentTime = currentTime;
      digitalWrite(INDENT_DETECTED_LED, HIGH);
      digitalWrite(NOT_ON_SIDEWALK_LED, LOW); // OFF sidewalk LED
    }
```

```
    //When indent isn't detected
    else {
      //Turn off indent found led
      digitalWrite(INDENT_DETECTED_LED, LOW);

      //If the indent hasn't been found for awhile, greater than
      //4 seconds
      if ((currentTime - lastIndentTime) > 4000) {
        //Turn on not on sidewalk LED
        digitalWrite(NOT_ON_SIDEWALK_LED, HIGH);
        Serial.println("NOT ON SIDEWALK!");
      }
      else {
        digitalWrite(INDENT_DETECTED_LED, LOW);
        Serial.println("Away from indent.");
      }
    }

  }
  else {
    Serial.println("OFF");
  }

  //digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  //digitalWrite(LED_BUILTIN, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

PROJECT THREE FINAL DELIVERABLE: PROJECT REFLECTION

The activities in this handout are intended to be completed by the end of the Project 3. You will apply what you learn in Design Communication Workshop #5 to complete this task.

Submission Details

Each Team Member: upload your reflection essay as a PDF to the Avenue Dropbox titled P3 Reflection using the MacID_P3_Reflection.pdf as naming convention

Grading of Reflection

Your reflection assignment is worth 1 mark of your total Project-3 grade (12.5%). Rubric is provided on Avenue to Learn.

If you need to review the content, go back to Design Communication Workshop #5 and/or go through the online reflecti0on module. Here is the link:
https://ecampusontario.pressbooks.pub/engineeringreflectiontoolkit/

Reflection Activity

Consider your experience with the design process as a first-year engineering student working on Project 3 over the past couple of months in ENG 1P13. After exploring the client's challenges and gaining insights, your team, decided to focus on one area to improve our client's daily life. You have defined the problem in a problem statement that included objectives, constraints, etc. Through this exploration, you performed a functional analysis that was used to come up with different alternative ways to solve the problem. Your team needed to make a decision between the different alternatives, and you tested your ideas for feasibility. You have been encouraged to iterate as you gained deeper insight and developed empathy for the client. Through the process of iteration, you have had the opportunity to improve upon your ideas. Engineers are continually iterating through the design process. Informed designers are involved in continual learning: learning by doing, learning from brainstorming and prototyping, learning by iteration and from feedback and failure, learning by noticing and troubleshooting, learning by drawing and dialoging about ideas, materials, and people. While iteration is an informal form of reflection, you will deepen your understanding of what you have learned through formal reflection. All of these emphasize the metacognitive and reflective practice aspects of learning through design (Lawson & Dorst, 2009; Crismond & Adams, 2012).

Part 1: What?

In this section you will describe a critical incident that you will be reflecting on as related to the "Generating/Testing ideas" and "Decision-making". For each of these steps of the design process:

In three to five sentences, identify and describe ONE critical incident, breakthrough or big thought-provoking moment that either challenged your assumptions, had a positive impact on you or validated your understanding of the design process. Here are some questions to consider.

Generating & Testing Ideas:

- How did you go about exploring ideas?
- How deeply did you explore your design options?
  - How much research?
  - Did you look into Biomimicry tools?
  - Did you consider any "What if?" questions in your explorations?
- Did you test your ideas?
- If yes, how did you test your ideas?
  - What were you trying to test (e.g., desirability, feasibility, etc.)?
  - What tool/ method did you use? (physical prototype, CAD model, etc.)
  - How much time did you spend on testing each idea?
  - How many ideas did you test?
  - How many prototypes did you make for testing each idea?
  - Did you test your ideas early on or waited until you had more details of the ideas?
- What was one challenge that you faced in the testing process of the design? (we encourage you to write more than one challenge). And What did you do to solve that challenge? (you can attach photos to explain your attempted solutions)
- From the results of our testing, one change we made to improve our design solution was … (add your response) and this change made our design solution better because … (add your response).

Response:

Decision Making:

- What happened during decision-making?
  - Where in the process, relative to the design process steps, did you make decisions?
  - What were the decisions about? Decisions could be about the process (e.g., how much searching of the design space was enough?) or about the design (e.g., which alternative to prototype).
  - How many options did you have to choose from?
  - How many criteria did you have to compare the options? How did you choose those criteria?
  - What tools did you use to make a decision?
- At what stage did you make a decision?
- When did this experience take place? Did you already have one final solution in mind or you were still exploring the ideas?
- What challenges did you face during decision-making process?

Response:

Part 2: "So What?"

In this section you will explore what you learned and describe why this incident matters to you.

In three to five sentences, discuss what you learned from this incident about idea generation, testing ideas, and decision making and that either surprised you, made you confront a misconception, or improved your understanding of the design process.

To help you think about this, consider the following:

- What was the outcome of early or late testing processes?
- Do you think delaying any of your decision-making may have improved the design?
- Could you have collected better observations or data that would have led to better decisions?
- Did you repeat your decision-making process at any other stage?

Response:

In two to three sentences, explain why these new insights are important to you.

Response:

Part 3: "Now What?"

In two to three sentences, discuss how you will integrate this new insight into future design projects, including next week prototyping and design review 2 where you still have a chance to improve your design. To help you think about this, consider the following:

- I learned that... (Express and important learning, not a statement of fact)
- This learning matters because... (Consider how this learning has value to you as an engineer)
- How will I apply my learning?
- How will I design differently next time?
- How will I deal with a similar situation in the future?
- Considering this learning, I will... (Set specific, assessable goals; consider benefits and challenges involved in this plan)

Response:

In two to three sentences, describe the possible benefits and challenges involved in your plan.

Response:

References:

Lawson, B., & Dorst, K. (2009). Design expertise. Oxford, UK: Architectural Press.

Crismond, D. P., & Adams, R. S. (2012). The informed design teaching and learning matrix. Journal of Engineering Education, 101 (4), 738-797.